



EAST TEXAS A&M
— UNIVERSITY —

CSCI 380.61E
WEB PROGRAMMING
AND INTERFACE DESIGN

OVERVIEW

- Review
- Fundamental **HTML** and **CSS** concepts?
- Review HTML and CSS **Syntax**
- Dynamic vs Static Websites
- **Examples**
- Questions and Discussion

REVISION QUESTION

What does IP Stand for?

- Internet Programming
- Internet Protocol
- Internal Package
- Internal Protocol

ANSWER

IP = Internet Protocol

REVISION QUESTION

What does TCP Stand for?

- Transfer Cache Practice
- Transmission Control Protocol
- Transmission Cache Protocol
- Test Cache Program

ANSWER

TCP = Transmission Control Protocol

REVISION QUESTION

UDP (User Datagram Protocol) is a communications protocol that is primarily used for establishing low-latency and loss-tolerating connections between applications on the internet.

- (a) True
- (b) False

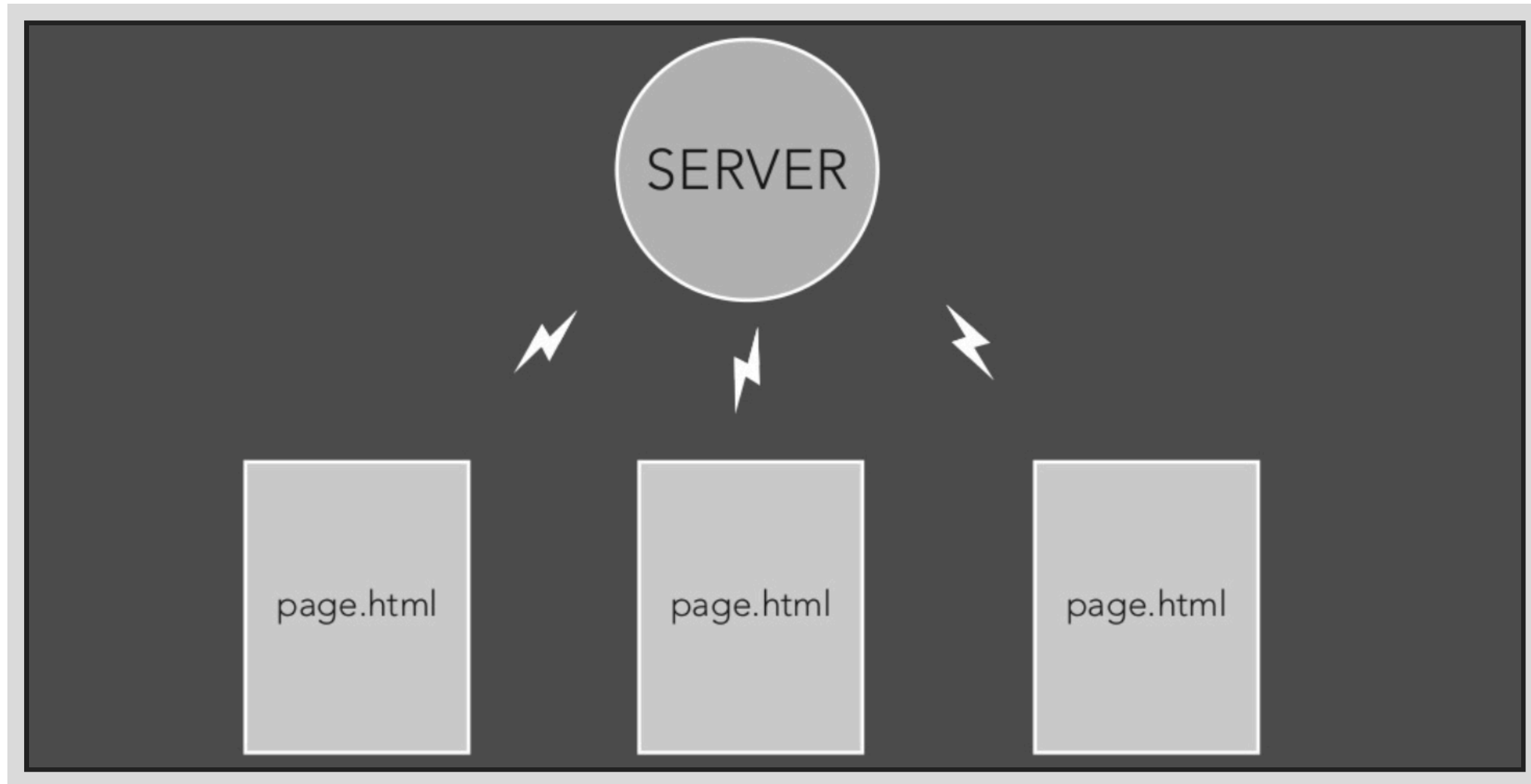
ANSWER

- (a) True

WHAT IS THE DIFFERENCE BETWEEN **STATIC** AND **DYNAMIC** WEB PAGES?

STATIC WEB PAGES

Static website pages are self-contained individual pages (or pages) sent to the browser from the server one-page-at-a-time



DYNAMIC WEB PAGES

Dynamic web content is built when the page is requested (by the user directly or programmatically while the user is on the page, e.g., Facebook updates)



HOW DO WE STRUCTURE CONTENT, PRESENTATION AND BEHAVIOUR?

- What are the **layers/structure** of web design?

BEHAVIOR

Javascript



PRESENTATION

CSS

Imagery



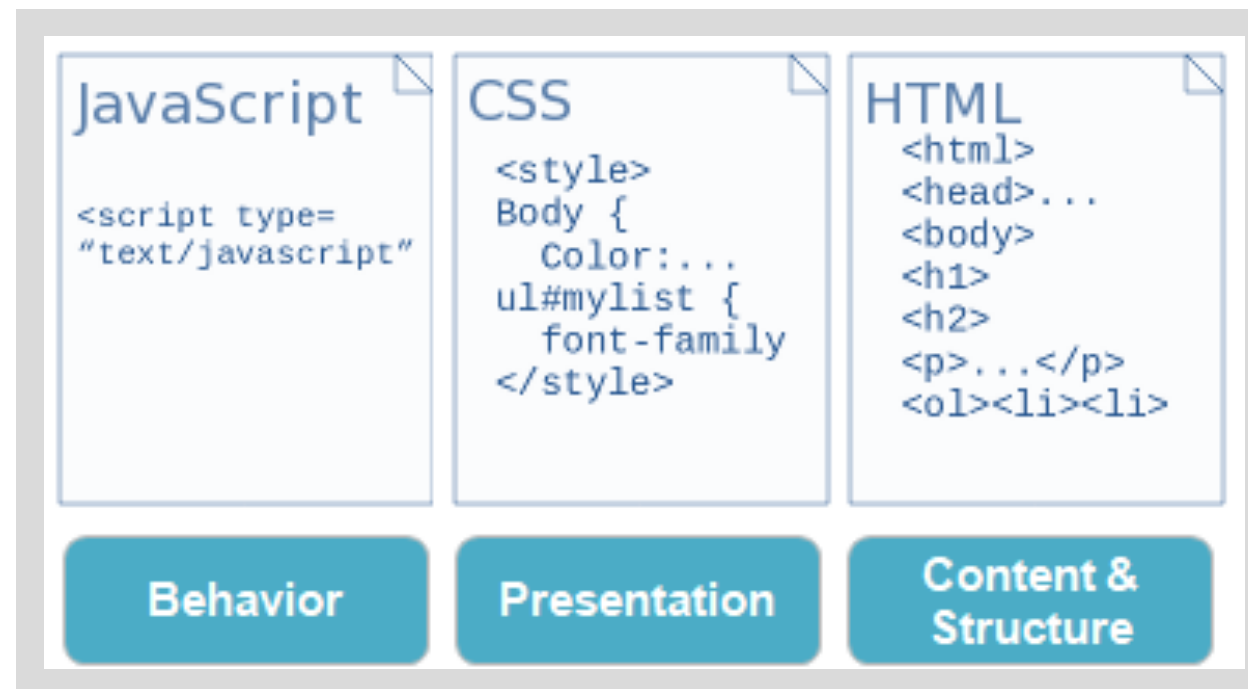
STRUCTURE

HTML markup

Site planning



FUNDAMENTALS OF A WEB PAGE



HTML



HYPER TEXT + MARKUP LANGUAGE

MARKUP LANGUAGE

- Set of **tags**
- Purpose is to group and describe **content**

```
<tag>Content</tag>
```

CAN YOU LIST THE ESSENTIAL HTML TAGS?



The essential element tags

Primary Structure

html

head

body

Head Elements

title

meta

link

Structural Elements (block)

p

br

h1 – h6

ul

ol

a

img

(div)

Formatting Elements (inline)

em

i

strong

b

q

blockquote

(span)

TAG ATTRIBUTES

- Additional information about the tag element

```
<html lang="en"></html>
```

- Tag element format

name="value"

REVISION QUESTION

Are attribute values case-sensitive?

- Yes
- No

ANSWER

Are **attribute** values case-sensitive?

- Yes

REVISION QUESTION

Basic syntax for HTML tags and attributes is:

- a. `<tag attribute=value> </tag>`
- b. `<tag attribute=value> <tag>`
- c. `<tag attribute="value"> <tag>`
- d. `<tag attribute="value"> </tag>`

ANSWER

Basic syntax for HTML tags and attributes is:

- d. `<tag attribute="value"> </tag>`

REVISION QUESTION

Choose the correct HTML tag for the largest heading

- a. <head>
- b. <heading>
- c. <h6>
- d. <h1>

ANSWER

Choose the correct HTML tag for the largest heading

- d. <h1>

REVISION QUESTION

What is the correct HTML tag for inserting a line break?

- a. `<lb>`
- b. `<break>`
- c. `<br>`

ANSWER

What is the correct HTML tag for inserting a line break?

- c. `<br>`

CSS



CASCADE + STYLE SHEET

STYLESHEET

- Rules defining **presentation**
- Targeted to specific **elements/tags** in the html

CASCADE

- Manages priorities and conflicts
- Multiple CSS rules
- Importances and **inheritance**

INHERITANCE

- Elements inherit style properties

HTML

relationship

<code><body></code>	→	parent of site
<code><div></code>	→	parent of ul and li, child of body
<code></code>	→	parent of li, child of div and body
<code></code>	→	child of ul, div, and body
<code></code>		
<code></div></code>		
<code></body></code>		

INHERITANCE

body

make the paragraph 16px, Verdana, red



p

make the paragraph blue



16px, Verdana, blue

EXAMPLE



CSS SYNTAX

- Great Control
- Override
- Specific

CSS CONTROL

- Three terms for describing style:

1. *CSS Rules*
2. *CSS Selectors*
3. *CSS Declaration*

CSS RULE

```
selector { property : value; }
```

- property:value is known as the **declaration**
- Every style has a **selector** and **declaration**
- property/value pair called a CSS Rule

CSS SELECTOR

```
body { font-family: Arial; }  
p {color: #112233; }  
a {color: red; }
```

- **selector** associates **css rules** with html elements

CSS SYNTAX

- CSS consists of a **selector** and a **declaration block**: The selector points to the **HTML element** you want to style
- **declaration block** contains one or **more** declarations separated by semicolons.
- Each declaration includes a CSS property **name and a value**, separated by a colon

CSS SELECTORS

- Type (tag/element)
- # ID
- . Class

TYPE (ELEMENT) SELECTORS

```
body {declaration}  
p {delcaration}  
h1 {delcaration}
```

- Simplest selector
- Targets html elements by name

ID SELECTORS

CSS

```
#logo {declaration}
```

HTML

```
<img id="logo" src="" alt="">
```

- ID is an **attribute** added to the html
- Reference ID in the CSS with a **hash**

CLASS SELECTORS

CSS

```
.cat {declaration}
```

HTML

```
<ul class="cat">
```

- class is a html **attribute** added to the markup language
- referenced in the css with a **period**

WHICH IS BETTER? ID OR CLASS?

ID VS CLASS

- Can only be **one ID**
- **Multiple classes**
- ID is more specific
- Element can have both ID and multiple classes

REVISION QUESTION

Which style is this?

```
<body style="background-color: blue">
```

- a. Inline style
- b. External style sheet
- c. External style stylist sheet
- d. None of the above

ANSWER

Which style is this?

```
<body style="background-color: blue">
```

- a. Inline style

QUESTION

The ____ property is used to set the color of the text.

- a. pallet
- b. color
- c. text-decoration
- d. design

ANSWER

The ____ property is used to set the color of the text.

- b. color

QUESTION

What does the following html/css code do?

```
<html><head>
  <style type="text/css"> p {font-size:32px;}
</head>
<body>
  This line displayed in default font size
  <p>Line formatted with element selector</p>
</body>
</html>
```

ANSWER

Displays Nothing

- Missing closing `</style>` tag

QUESTION

What does the following HTML/CSS do?

```
<html><head>
  <style type="text/css">
    #heading {color: #0000ff}
  </style>
</head>
<body>
  <p id="heading">Hello Hello</p>
  <p id="hello">What is happening here?</p>
</body>
</html>
```

ANSWER

Blue `Hello Hello`

- Associated with a **valid** selector ID

HOW DO YOU VALIDATE YOUR HTML AND CSS?

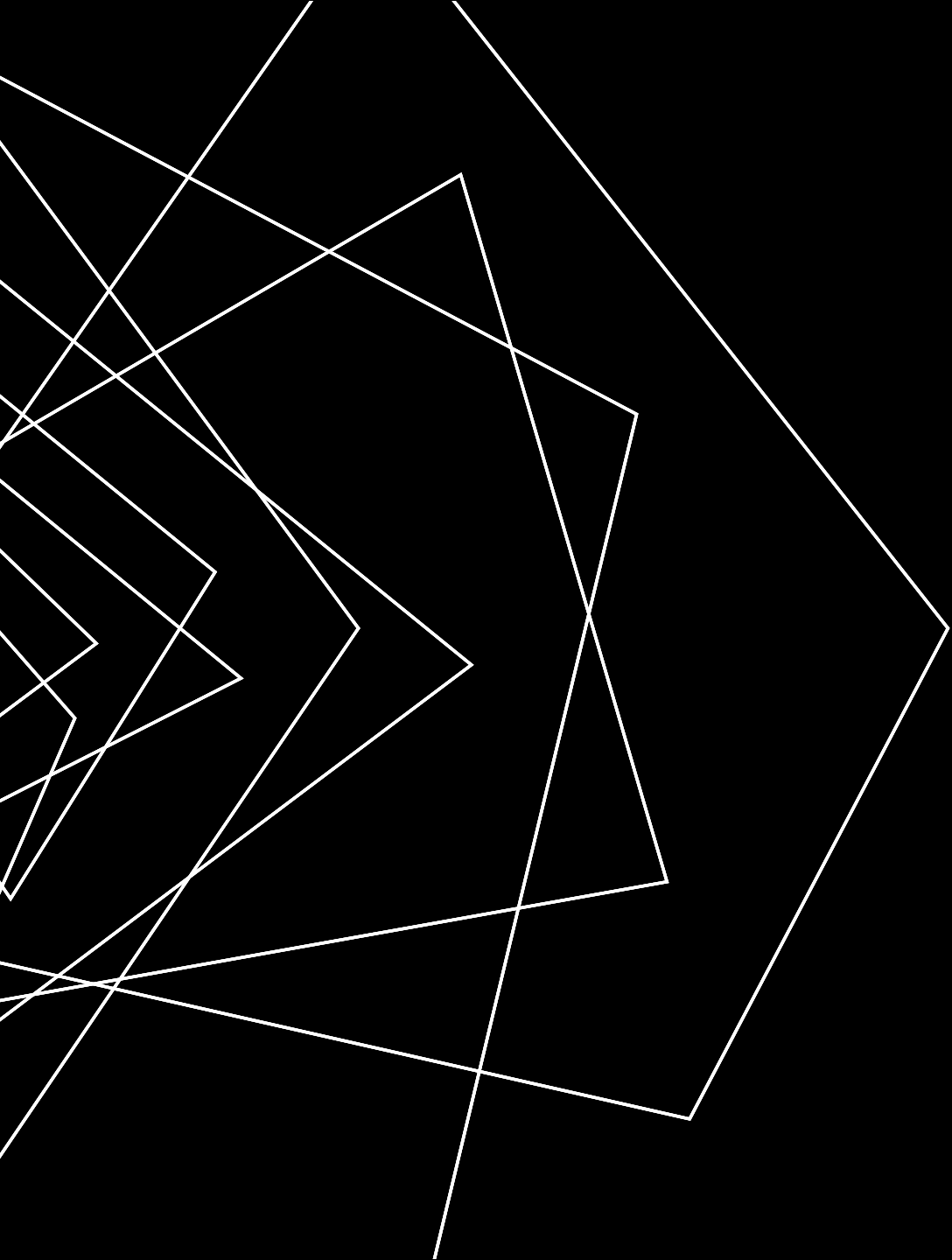


HOW DO YOU VALIDATE YOUR HTML AND CSS?

- <https://validator.w3.org>
- <http://jigsaw.w3.org/css-validator/>
- <https://validator.w3.org/checklink>

SUMMARY

- Understand **fundamental** HTML and CSS concepts
- Essential **syntax**
- Layers/management (HTML, CSS and JavaScript)
- Review (small taste)



THANK YOU

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